MILITARY SPECIFICATION

DISPLAY, IMAGE 11744684

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the general requirements for the Display, Image of the Target Selector Group, Fire Control, XM-10 (Vulcan), hereafter referred to as the Display.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATION

Military

MIL-F-13926	Fire Control Material: General Specification Governing the Manufacture and inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Material, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling, Procedure and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Term and Definitions
MIL-STD-171	Systems for Preparation, Painting and Finishing Metal and wood Surfaces
MIL-STD-275	printed Wiring for Electronic Equipment

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MIL-STD-454	Standards, General Requirements for Electronic Equipment
MIL-STD-461A	EMI Characteristics, Requirements for Equipment
MIL-STD-462	EMI Characteristics, Measurement of
MIL-STD-810	Environmental Test Methods

Federal

FED-STD-595A	Colors

DRAWINGS

Military

11744635	Identification Plate

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 ITem identification. - The Display is a high resolution, high brightness cathode ray tube unit that accepts electrical signals corresponding to an aircraft image and projects this image through a set of optics for viewing by personnel utilizing the target selector group, Fire Control, XM-10 (Vulcan).

3.1.1 Preproduction sample(s). - Preproduction sample(s) shall be given qualification tests prior to regular production to demonstrate the adequacy and suitability of the contractor's process and procedures in achieving the performance that is inherent in the design.

3.2 Reliability. - The display shall be designed for 5,000 hours mean-time-between-failure MTBF).

3.3 Characteristics.

3.3.1 Performance. - The Display shall be manufactured as specified in the applicable specification drawing 11744684.

3.3.1.1 Environmental.

3.3.1.1.1 Temperature. - The Display shall be capable of being stored at +160°F and -67°F and operated after being temperature stabilized

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at +140°F and -4°F. Subsequent to such exposure the Display shall not have been damaged or adversely affected and shall meet electrical performance requirements of this specification at standard room ambient conditions.

3.3.1.1.2 Humidity.

3.3.1.1.2.1 Operating.- The display shall be capable of meeting the performance requirements of this specification while exposed to an atmosphere with a relative humidity (R.H) of 94 ± 4 percent at a temperature of 86°F.

3.3.1.1.2.2 Non-operating. - The Display shall be capable Of Withstanding 240 hours of continuous exposure to a relative humidity (RH) of 94 ±4 percent at temperature cycling of from 73°F to 140°F.

3.3.1.1.3 Shock. The Display shall be capable of withstanding three 10G 11 milliseconds shock in each of its mutually perpendicular axis for a total of 18 shocks.

3.3.1.1.4 vibration.- The Display shall be capable of Withstanding a vibration having a double amplitude of 0.076 in. and acceleration levels of from 2 to 4G applied logarithmically over the 5 to 500 Hz frequency range for a 15 minute time duration.

3.3.1.1.5 Salt fog. - The Display shall be capable of withstanding forty-eight (48) hours of continuous exposure to a 5% salt solution with a specific gravity of 1.027 to 1.034, a PH of 6.5 to 7.2 and a temperature of 95°F.

3.3.1.1.6 Sand and Dust.- The Display shall be capable of operation after thirty six hours of exposure to wind driven sand and dust.

3.3.1.1.7 Rain. - The Display shall be capable of operation after exposure to a 2in/hr, 20Mph wind driven rain. The water temperature shall be between 60 and 90°F.

3.3.1.1.6 Altitude. - The Display shall be capable of operating during exposure at an altitude of 10,000 ft. after exposure at an altitude of 50,000 ft.

3.3.1.1.9 Fungus. - The Display shall be capable of withstanding a twenty-eight (28) day fungus exposure to a composite spore suspension.

3.3.1.1.10 EMI. - The Display shall meet the requirements of MIL-STD-461A subtest RE02.

3.4 Design and Construction

3.4.1 Composition.- To the maximum extent practical, the equipment shall consist of solid-state circuitry mounted on printed circuit cards.

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3.4.2 Printed Circuit Cards.- All printed circuit cards used shall be designed and manufactured in accordance with MIL-STD-275.

3.4.3 Exterior Surfaces.- On exterior surface finish 21.3 of MIL-STD-171 using primer TT-P-1757 color Y. Bake primer and top coat color Olive Drab No. 24087 of FED-STD-595a.

3.5 Design and Performance.

3.5.1 Weight. The Display shall weigh amaximum of 29 lbs.

3.5.2 Input Power.- The Display shall operate with no degradation over the voltage range of 24-30 VDC at 75 watts max.

3.5.3 Electrical Connections.- Electrical connections shall be made through a 10-107924-20S (Bendix) connector. A threaded protective cap & chain shall be provided with the connector.

TABLE I

Conector Type - 10-107924-20S(Bendix)

PIN	SIGNAL
A	Vertical position
B	Vertical return
C	Blanking
D	Blanking return
E	+28 volts
F	+28 volts return
G	Spare
H	Horizontal position
J	Horizontal return
K	Spare
L	Spare

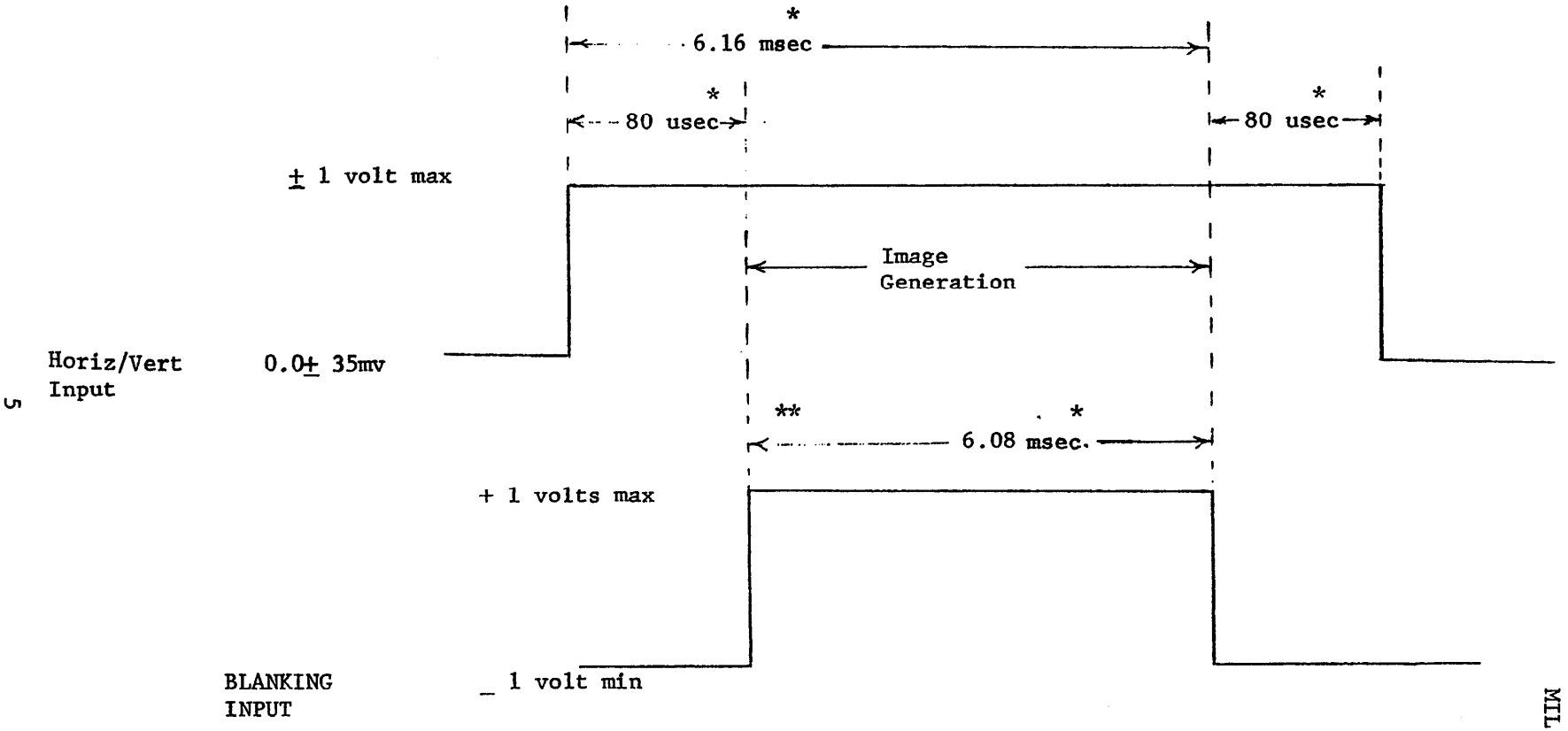
3.5.3.1 Wiring.- The connector shall be wired per Table 1.

3.5.3.2 Input format. - The input format shall be in accordance with Figure 1.

3.5.3.3 Input impedance. - Input impedance for the horizontal and vertical inputs shall be 75 ohms ± 0.1%. Input impedance for the blanking input shall be 5K ohms minimum.

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DISPLAY INPUT FORMAT



* All timing tolerances are ± 5%

** The start of the Image Generation interval is time coincident with unblanking period

FIGURE 1

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MIL-D-48437A(MU)

3.5.4 Controls and Indicators.

3.5.4.1 Brightness.- A rotary brightness control shall be provided. Clockwise rotation shall increase the brightness linearly with the control position.

3.5.4.2 Contrast.- A rotary-contrast control shall be provided. Clockwise rotation shall increase the contrast linearly with the control angular position.

3.5.4.3 Standby/On.- A Standby/On switch shall be provided such that in the standby position the display is blanked. Upon selection of "On" the correct symbolic presentation shall appear after a 2 minute maximum warm-up time$_c$

3.5.4.4 Self Test.- A self test shall be included to provide to dynamic check of the major display subassembly. The self test switch shall have the following positions:

a. off
b. LVPS
c. MON
d. TEMP
e. DEFL
f. VIDEO
g. HVPS

When the self test switch is in any position other than "OFF" a 2 inch diameter circle shall be displayed in the center of the crt a frequency of 25 KHz.

3.5.4.4.1 Go- No Go Test.- A go - no go indicator shall be provided to depict the operational status of the Display when the self test switch is cycled through the subassembly test positions of paragraph 3.504.4. A green light shall indicate that the subassembly under test is operational; the light extinguished shall indicate an inoperative subassembly.

3.5.4.4.2 Image Display (ID) Failure. - An ID fail light shall be incorporated to indicate that one or more of the self test checks has failed. The light shall be yellow to indicate an inoperative subassembly. When the ID fail light is "on" the High Voltage power supply and deflection subassembly shall be placed in a standby status. Performance of the self test shall indicate the faulty subassembly on the go - no go indicator.

3.5.5 Display Characteristics.

3.5.5.1 Size.- The active viewing area shall be 5" vertical X 7" horizontal. The CRT radius shall be 60" with a face plate thickness of 0.3".

3.5.5.2 Phosphor.- The phosphor shall be P31.

3.5.5.3 Line Definition.- With the brightness control set at maximum the Display shall have a maximum line width of 0.015 inches in the central 90% of the viewing area measured at the 50% line brightness points.

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3.5.5.4 Refresh Rate. - The Display shall have a refresh rate of 60 Hz ± 5%.

3.5.5.5 Writing.- The display writing speed shall be defined by Figure 2 for a basic sine wave input. All timing tolerances shall be ±5%.

3.5.5.5.1 Blanking. - The blanking signal bandwidth shall be 2 MHz minimum.

3.5.5.6 Positional Stability. - Screen center (O volt inputs on pins A and H of Table I) shall not change by more than± .035 inches during all testing of the Image Display.

3.5.5.6.1 Deflection Accuracy.- The display shall be aligned in accordance with Table II to an accuracy of ±.010″ at screen center. The accuracy tolerance shall increase linearly to a nuiximum of ± .020″ at full screen deflection.

3.5.5.6.2 Environmental Accuracy.- The deflection accuracy toleranes shall increase under environmental testing to an accuracy of ±l.020″ at screen center. The accuracy tolerance shall increase linearly to a maximum of ±0.030" at full screen deflections

3.5.5.6.3 Resolution.- The horizontal and vertical resolution for the displayed symbology shall be 0.007 inches.

3.5.5.6.4 Deflection Convention.- With the Display control surface at the top a positive input on the horizontal and vertical input lines shall cause the beam to move to the left and up from screen center.

3.5.6 Brightness.- With 1" diameter circle of 25 KHz frequency and an unblinking time of 0.04 milliseconds and a 60Hz refresh rate the Display shall have a brightness of 700 foot lamberts measured between the 50% points of the circle line width.

3.5.7 CRT Protection.- Deflection failure safety circuits shall be provided such that a loss of beam deflection shall be sensed and the CRT electron beam current automatically cut off and the self test circuitry triggered.

3.5.7.1 Input Protection.- The CRT shall not be damaged (including bum marks) by the application of power in the absence of deflection or blanking input signals.

3.5.8 Stray Reflections.- Stray reflections shall be minimized by the use of high efficiency anti-reflective coatings on the CRT face.

3.5.9 Sealing.- The CRT face shall be sealed from the Display electronics.

3.5.10 Touch Temperature.- The maximum touch temperature allowable on the control panel shall be less than 25°F above the ambient temperature.

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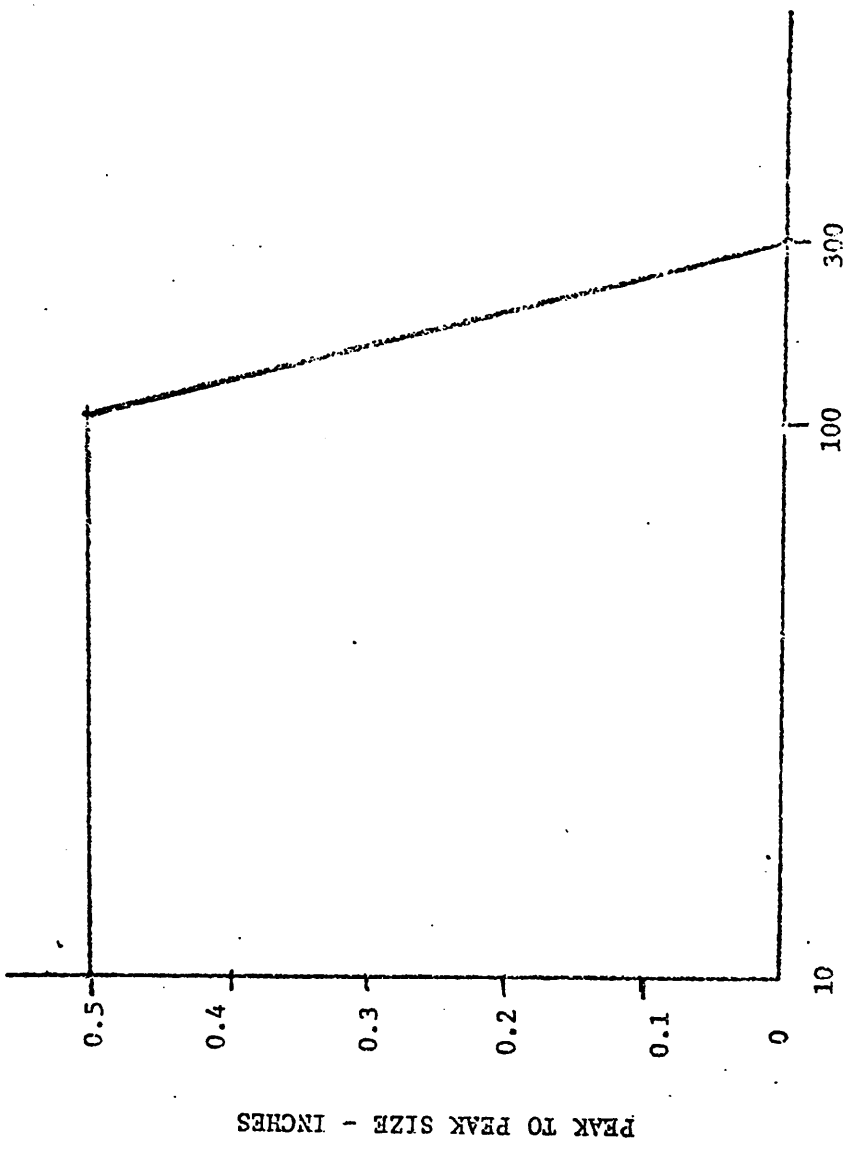


FIGURE 2

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TABLE II

Display postition $d = 9.030 \tan 20E = 3.121 \tan^3 20E$

where d = spot position in inches from screen center

E = horizontal or vertical input voltage

E (volts)	d (inches)
0	0
± .1	± 0.315
± .2	± 0.630
± .3	± 0.945
± .4	± 1.260
± .5	± 1.575
± .6	± 1.889
± .7	± 2.203
± .8	± 2.516
± .9	± 2.827
±1.0	± 3.136

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3.5.11 Thermal. - The Display shall be cooled through conduction and radiation.only.

3.5.12 Safety.- The design of high voltage power supplies and associated wiring and connector shall be such that no safety hazard is presented to maintenance personel. High voltage areas of the Display shall be suitably marked.

3.6 Reliability Run-in.- The Display shall be subjected to a 50 hour run-in period (excluding downtime) which includes the operation at temperature extremes. The 50 hour run-in period shall start after the Display has satisfactorily met the performance requirements of this specification. The time required to conduct the vibration test shall not be included as part of the run-in period. The Display shall continuously meet the requirement of this specification for the entire 50 hour run-in period.

3.6.1 Vibration. - The Display shall not be damaged after being subjected to simple harmonic motion for a period of 30 minutes.

3.6.2 Temperature extremes.- The Display shall be capable of operating without damage for 4 hours at -4°F and for 4 hours at 125°F.

3.6.3 Falures during Reliability Run-in.- Replacement of defective components and correction of workmanship defects may be made during run-in under the following conditions:

a. Any part, module component or subassembly that is used as a replacement shall have previously passed environmental tests consistent with the reliability run-in environmental requirements of the Display.

b. Any replacement part shall have been exposed for at least one (1) hour at the extreme temperature being run immediately previous to being used as a replacement for any failure occuring during a temperature extreme.

c. After acorrection has been accomplished, at least one (1) hour of operation with a minimum of three (3) tests to the part, module, component or subassembly requirements shall be successfully accomplished at the temperature at which the failure occurred.

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4. QUALITY ASSURANCE PROVISION

4.1 General.

4.1.1 Responsibility for inspection. - Unless otherwise specified in the contract, the supplier is responsible for the performance of all test and inspection requirements specified herein. Except as otherwise specified in-the contract the supplier will specify facilities suitable for the performance of the test and inspection, unless disapproved by the Government. The Government reserves the right to perform any of the tests and inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1.1 quality program requirements. - The contractor shall provide and maintain a quality program acceptable to the Government for supplies and services covered by this specification.

4.1.1.1.1 Quality plan. - A quality plan, applicable to the requirements of this specification shall be submitted to the contracting officer for approval prior to first article inspection. If first article inspection is not required the quality plan shall be submitted for approval prior to quality conformance inspection. The quality plan shall reflect the following:

a. Flow of material from inspection on receipt throuth all manufacturing processes to final shipment.

b. Process and inspection points.

c. Reference to process and inspection procedures.

d. Differentiation between in plant inspections and subcontracted inspections for subassemblies and higher assemblies.

Any changes to the approved plan shall be submitted to the contracting officer for approval prior to use.

4.1.2 Government verification. - All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered product to assure compliance with all requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made and until conformance of product to prescribed criteria has been demonstrated.

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4.1.3 Test procedures. - The contractor shall submit written test procedures including test equipment and tolerance limits to be used for first article inspection and quality conformance inspection (see 6.1.2). Each test shall be identified in accordance with 4.2 as to classification.

4.1.3.1 Procedures for inspections, tests, and demonstrations shall contain step by step information which will enable the operator to properly perform the requirements of this specification. The method of examination shall be clearly defined.

4.1.3.2 The procedures shall contain a list of test equipment identified by name and model. Special test equipment shall also be listed and identified.

4.1.3.3 Data sheets shall be included in the procedures. Conformance parameters shall be clearly indicated with tolerance limits. Whenever qualitative measurements are made, the result shall be shown by quantitative data.

4.2 Classification of inspection. - The method of examination and testing of the equipment fall within the following classifications:

a. First Article Inspection (see 4.3).

b. Quality Conformance Inspection (see 4.4).

(1) Production Inspection (see 4.4.1).

(2) Production Control Inspection (see 4.4.2).

(3) Environmental Tests (see 4.5).

c. Reliability Testing (see 4.7).

4.3 First article inspection.- Unless otherwise specified, one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification.

4.3.1 First article inspection reort. - A first article inspection report shall be prepared containing all the results of the first article inspection.

4.4 Quality conformance inspection.

4.4.1 Production inspection. - Production inspection shall be made on every display offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material.

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4.4.2 Production control inspection. - Three displays shall be selected at random as production control samples from each 50 produced. The control samples shall meet the requirements and tests of Table III.

TABLE III

TEST AND REQUIREMENTS

TEST	REQUIREMENT	TEST PROCEDURES
Design and Performance	3.5	
Temperature	3.3.1.1.1	4.5.1
Humidity	3.3.1.1.2	4.5.2
Shock	3.3.1.1.3	4.5.3
Vibration	3.3.1.1.4	4.5.4
Salt Fog	3.3.1.1.5	4.5.5
Sand and Dust	3.3.1.1.6	4.5.6
Rain	3.3.1.1.7	4.5.7
Altitude	3.3.1.1.8	4.5.8
Fungus	3.3.1.1.9	4.5.9
Design and Construction	3.4	
Reliability Run-in	3.7	4.8

4.4.3 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements. acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.2.2).

4.4.4 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use till not unduly interfere with the contractor's delivery schedule.

4.4.4.1 Contractor designed equipment. - The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in the detail specification and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerance used in the design test equipment shall be so selected and controlled as to ensure that the test equipment till reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerances limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.4.2 Test equipment. - Test equipment used to test assemblies for verification of Section 3 requirements shall be as specified in the test procedure.

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4.4.5 Examinations.- Unless otherwise specified, components and subassemblies shall be inspected in accordance with the "Inspection provisions for components and subassemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations to verify requirements of 3.3 shall be performed. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.4.6 Functional tests. - Detail functional requirements shall be inspected on a 100 percent basis. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a. Temperature	+50°F to +90°F
b. Relative humidity	90 percent RH max.
c. Pressure	28 to 32 inches of Hg

4.5 Environmental test methods and Procedures.

4.5.1 Temperature storage.

a. High temperature. - This test shall be conducted in accordance with Procedure I, Method 501 of MIL-STD-810B.

b. Low temperature.- This test shall be conducted in accordance with Procedure 1, Method 502, of MIL-STD-810B.

4.5.2 Humidity.- The Display shall be subjected to the humidity test as outlined in MIL-STD-810, Method 507, Procedure II, except that performance test to be conducted during the exposure shall be conducted during the 1st, 3rd and 5th cycles; subsequently the display shall not be damaged or adversely affected and shall meet the performance requirements of this specification.

4.5.3 Shock. - The Display shall be subjected to three 10G, 11 millisecond duration, shocks in each of the three mutually perpendicular directions for a total of 18 shocks as outlined in MIL-STD-810, Procedure I. The Display shall be energized and a test pattern observed on a monitor. Subsequent to the shock test, the display shall not have been damaged or adversely affected and shall meet the performance requirements of this specification.

4.5.4 Vibration. - The Display shall be subjected to the vibration as specified in MIL-STD-810, Method 514, Procedure VIII except that the displacement and frequency shall be as follows:

GRAVITY	FREQUENCY (Hz)
0.076 in. double	5 - 23
2G	23- 40
4G	40- 500

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Subsequent to the vibration test, the Display shall not have been damaged or adversely affected and shall meet the performance requirements of this specification

4.5.5 Salt fog. - The Display shall be subjected to the salt fog test as specified in MI-STD-810, Method 509. Subsequent to the exposure the Display shall not be damaged or adversely affected and-shall meet the performance requirements of this specification.

4.5.6 Sand and dust. - The Display shall be subjected to the sand and dust test as specified in MIL-STD-810, Method 510. Subsequent to the exposure the Display shall not be damaged or adversely affected and shall meet the performance requirements of this specification.

4.5.7 Rain. - The Display shall be subjected to the rain test as specified in MIL-STD-810, Method 506. The Display will not be operatied during the test, but subsequently it shall not have been damaged or adversely affected and shall meet the performance requirements of this specification.

4.5.8 Altitude. - The Display shall be subjected to the altitude as specified in MIL-STD-810, Method 500, Procedure I. Subsequent to the exposure, the Display shall not have been damaged or adversely affected and shall meet the performance requirements of this specifications

4.5.9 Fungus. - The Display shall be subjected to the Fungus Test as specified in MIL-STD-810, Method 508, for 28 days. Subsequent to the exposure the Display shall not have been damaged or adversely affected and shall meet the performance requirements of this specification.

4.5.10 Non-conforming environmental sample units. - If a sample unit fails the test specified in 4.5 the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials, processes. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.6 Electromagnetic compatibility.- This test shall be conducted in accordance with MIL-STD-462, Method RE02.

4.7 Reliability demonstration. - A reliability demonstration test shall be conducted as part of the First Article Test to assure compliance with 3.5 at a 90% confidence level. The systems shall operate at least 4 hours during exposure to the operating environments of 4.3.4 and after the non-operating environment exposures. Additional hours shall be accumulated at 75 ± 15°F until the total test operating hours meet the accept/reject criteria in Table V. The performance requirements of 4.4 shall be performed at least once each operating hour.

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TABLE V

TOTAL OPERATING HOURS	RELIABILITY FAILURES ACCEPT NO.	REJECT NO.
10400	0	2
	1	2

4.7.1 Data collection. - Data collected till consist of all equipment malfunctions, number of hours completed when malfunction occurred, nature and cause of malfunction, time required to correct malfunction, required corrective action, item serial number and environmental exposure.

4.7.2 Failure definition. - A failure is defined as any malfunction which renders the system inoperative or operative outside the limits of this specification. In determination of chargeable reliability failures the following clarification shall apply:

a. If upon failure analysis (to determine the cause of the malfunction) an assignable cause (related to manufacturing process, procedure, basic design weakness, etc.) is determined as the basis of the malfunction, the malfunction will not be chargeable as a reliability failure. However, correction to the manufacturing process or procedure (et al) shall be accomplished.

b. If upon failure analysis no assignable cause is evident and the malfunction is considered random in nature, the malfunction shall be a “chargeable” reliability failure and as such must be considered in the above demonstratio test criteria. The decisions as to chargeable reliability failure must be co-ordinated with the procuring activity, Frankford Arsenal Reliability and Maintainability Engineering Branch, SARFA-QAF-R.

4.7.3 System interruption. - A system interruption is any system malfunction for which the system is operable within the limits set forth in this specification, but requires maintenance to correct the malfunction. Although a system interruption does not penalize the reliability, it may constitute a failure of some major subassembly or component for which system acceptance cannot be made until that characteristic or component which caused the system interruption has been screened.

4.8 Reliability Run-in. - Subsequent to the successful performance requirements of this specification the display shall be subjected to 50 hours of Run-in time. The Run-in shall exclude any down-time necessary for repairs. It shall include vibration, followed by at least four hours at -4°F and at least four hours at +125°F.

4.8.1 Vibration Run-in. - The vibration test shall be performed as specified in MIL-STD-810, Method 514.1, Procedure IX, Part I. Subsequent to the vibration test, the Display shall not have been damaged or adversely affected and shall meet the performance requirements of this specification.

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4.8.2 Temperature Run-in.- The Display shall be placed in a chamber and the temperature gradually raised at a rate not to exceed 2°F per minute until thermally stabilized at plus 125°F. Five minutes after power is applied, a complete test to the performance requirements of paragraph 4.3.12 shall commence. A second complete test to the requirements of paragraph 4.3.12 shall be accomplished during the middle of the four hour period and a theird test to the performance requirements of paragraph 4.3.12 shall be accomplished during the last part of the hour test. Repeat the above tests at a temperature of -4°F. Subsequent to the above temperature extremes (plus 125° and -4°F) the Display shall be not have been damaged or adversely affected and shall meet the performance requirements of this specification.

4.8.3 Failures. - Failures tiring Reliability Run-in shall not be cause for rejection, but shall be corrected as outlined in 3.6.3.

5. PREPARATION FOR DELIVERY

5.1 General.- Packaging and marking shall be as specified in the contract.

6. NOTES

6.1 Intended Use.- The Display, Image commonly referred to as the Display, is one of the subassemblies of the XM-10 Vulcan Gunner Tracking Evaluator which is used to evaluate the proficiency of Vulcan gunners.

6.2 Ordering Data.

6.2.1 Procurement requirements.- Procurement documents should specify the following:

a. Title, number and date of this specification~
b. Instructions for arranging of examination, test and approval of First Article (see 4.4)

6.2.2 Contract data requirements.- Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423). The data required by this specification includes, but is not restricted to the following:

(a) Reliability demonstration (see 3.2.1).
(b) Quality plan (4.1.1.1.1).
(c) Test procedures (see 4.1.3).
(d) Reliability demonstration (see 4.7).

6.3 Instruction for samles.

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6.3.1 Instruction for Preproduction samPle. - Production samples shall be as specified in the contract.

6.3.2 Tests. - Any change to control tests specified in 4.4.2 is to be so specified in the contract.

Custodian:

Army-MU

Preparing activity:

Army - MU

Project No. FSC-1240A556

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